MATH 243: Discussion Worksheet 11 **Answer Key**

Topics: 15.7 Triple Integrals in Cylindrical Coordinates; 15.8 Triple Integrals in Spherical Coordinates

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1. Rewrite each equation in cylindrical coordinates and identify the surface.

   (a) $x^2 + y^2 = 4$

   $r = 2$, a circular cylinder of radius 2 centered on the z-axis.

   (b) $z = 4 - x^2 - y^2$

   $z = 4 - r^2$, a circular paraboloid opening downward, with vertex $(0, 0, 4)$.

   (c) $x^2 + y^2 + z^2 = 9$

   $r^2 + z^2 = 9$, a sphere centered at the origin with radius 3.

2. Find the volume of each solid by computing $\iiint_E dV$ using cylindrical coordinates.

   (a) The solid in the first octant that lies between the cylinders $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$, and the planes $z = 0$ and $z = 4$.

   The solid projects onto the xy-plane as the part of the annulus $1 \le r \le 2$ in the first quadrant. Since the solid lies in the first octant, $0 \le \theta \le \frac{\pi}{2}$. Also, $0 \le z \le 4$. Thus,

   $$E = \left\{(r, \theta, z) : 0 \le \theta \le \frac{\pi}{2},\ 1 \le r \le 2,\ 0 \le z \le 4\right\}.$$

   The volume is

   $$\begin{aligned}
   \mathrm{Vol}(E) &= \iiint_E dV \\
   &= \int_0^{\pi/2} \int_1^2 \int_0^4 r \, dz \, dr \, d\theta \\
   &= \int_0^{\pi/2} \int_1^2 4r \, dr \, d\theta \\
   &= \int_0^{\pi/2} \left[2r^2\right]_1^2 \, d\theta \\
   &= \int_0^{\pi/2} 6 \, d\theta \\
   &= 3\pi.
   \end{aligned}$$

   (b) The solid that lies inside the cylinder $x^2 + y^2 = 4$, above the plane $z = 0$, and below the plane $z = 3 + x$.

In cylindrical coordinates, the cylinder becomes $r = 2$ and the upper plane becomes $z = 3 + r\cos\theta$. Since $-2 \le r\cos\theta \le 2$, the upper plane lies above $z = 0$ throughout the disk $r \le 2$. Thus,

$$E = \{(r, \theta, z) : 0 \le \theta \le 2\pi,\ 0 \le r \le 2,\ 0 \le z \le 3 + r\cos\theta\}.$$

The volume is

$$\begin{aligned}
\mathrm{Vol}(E) &= \iiint_E dV \\
&= \int_0^{2\pi}\int_0^2\int_0^{3+r\cos\theta} r\,dz\,dr\,d\theta \\
&= \int_0^{2\pi}\int_0^2 (3r + r^2\cos\theta)\,dr\,d\theta \\
&= \int_0^{2\pi}\left[\frac{3r^2}{2} + \frac{r^3}{3}\cos\theta\right]_0^2 d\theta \\
&= \int_0^{2\pi}\left(6 + \frac{8}{3}\cos\theta\right) d\theta \\
&= \left[6\theta + \frac{8}{3}\sin\theta\right]_0^{2\pi} \\
&= 12\pi.
\end{aligned}$$

(c) The solid enclosed by the cylinder $x^2 + z^2 = 4$ and the planes $y = -1$ and $y + z = 4$.

Since the cylinder is $x^2 + z^2 = 4$, we use cylindrical coordinates about the y-axis:

$$x = r\cos\theta, \qquad z = r\sin\theta, \qquad 0 \le r \le 2, \qquad 0 \le \theta \le 2\pi.$$

Here r is the distance from the y-axis. The two planes give $-1 \le y \le 4 - r\sin\theta$. The upper bound is always greater than the lower bound, since $r \le 2$. Thus,

$$\begin{aligned}
\mathrm{Vol}(E) &= \int_0^{2\pi}\int_0^2\int_{-1}^{4-r\sin\theta} r\,dy\,dr\,d\theta \\
&= \int_0^{2\pi}\int_0^2 (5r - r^2\sin\theta)\,dr\,d\theta \\
&= \int_0^{2\pi}\left[\frac{5r^2}{2} - \frac{r^3}{3}\sin\theta\right]_0^2 d\theta \\
&= \int_0^{2\pi}\left(10 - \frac{8}{3}\sin\theta\right) d\theta \\
&= \left[10\theta + \frac{8}{3}\cos\theta\right]_0^{2\pi} \\
&= 20\pi.
\end{aligned}$$

3. Set up a triple integral using cylindrical coordinates that gives the volume of the solid inside the sphere $x^2 + y^2 + z^2 = 2$ and above the paraboloid $z = x^2 + y^2$. *Do not evaluate.*

In cylindrical coordinates, the sphere is $r^2 + z^2 = 2$, and the paraboloid is $z = r^2$. The upper surface is the upper half of the sphere, $z = \sqrt{2 - r^2}$. The two surfaces intersect when

$$r^2 + (r^2)^2 = 2 \quad \Longleftrightarrow \quad (r^2 - 1)(r^2 + 2) = 0,$$

so $r = 1$. The projection onto the xy-plane is therefore the disk $0 \le r \le 1$. Thus,

$$E = \left\{(r, \theta, z) : 0 \le \theta \le 2\pi,\ 0 \le r \le 1,\ r^2 \le z \le \sqrt{2 - r^2}\right\},$$

and the required integral is

$$\mathrm{Vol}(E) = \int_0^{2\pi} \int_0^1 \int_{r^2}^{\sqrt{2-r^2}} r \, dz \, dr \, d\theta.$$

4. Rewrite the following iterated integral using cylindrical coordinates. *Do not evaluate.*

$$\int_{-3}^{3} \int_0^{\sqrt{9-x^2}} \int_0^{9-x^2-y^2} \sqrt{x^2 + y^2} \, dz \, dy \, dx$$

The bounds for x and y describe the upper half of the disk $x^2 + y^2 \le 9$. Thus $0 \le \theta \le \pi$ and $0 \le r \le 3$. The solid lies above the plane $z = 0$ and below the paraboloid $z = 9 - r^2$. Also, $\sqrt{x^2 + y^2} = r$. Including the factor r from the change of coordinates, the integral becomes

$$\int_0^{\pi} \int_0^3 \int_0^{9-r^2} r^2 \, dz \, dr \, d\theta.$$

5. Using spherical coordinates, evaluate the triple integral

$$\iiint_E (x^2 + y^2 + z^2) \, dV,$$

where E is the solid region between the spheres $x^2 + y^2 + z^2 = 4$ and $x^2 + y^2 + z^2 = 9$.

The two spheres become $\rho = 2$ and $\rho = 3$ in spherical coordinates. The solid region is given by

$$E = \{(\rho, \theta, \phi) : 0 \le \phi \le \pi,\ 0 \le \theta \le 2\pi,\ 2 \le \rho \le 3\}.$$

Since $x^2 + y^2 + z^2 = \rho^2$, we have

$$\begin{aligned}
\iiint_E (x^2 + y^2 + z^2) \, dV &= \int_0^{\pi} \int_0^{2\pi} \int_2^3 \rho^4 \sin\phi \, d\rho \, d\theta \, d\phi \\
&= \int_0^{\pi} \int_0^{2\pi} \frac{3^5 - 2^5}{5} \sin\phi \, d\theta \, d\phi \\
&= \frac{211}{5}(2\pi) \int_0^{\pi} \sin\phi \, d\phi \\
&= \frac{422\pi}{5} \left[-\cos\phi\right]_0^{\pi} \\
&= \frac{844\pi}{5}.
\end{aligned}$$

6. Find the volume of the solid that lies within the sphere $x^2 + y^2 + z^2 = 4$, above the xy-plane, and below the cone $z = \sqrt{x^2 + y^2}$. Use spherical coordinates.

In spherical coordinates, the sphere becomes $\rho = 2$. Since $z = \sqrt{x^2 + y^2}$ becomes $\rho\cos\phi = \rho\sin\phi$, the cone is described by $\phi = \frac{\pi}{4}$. Being above the xy-plane gives $0 \le \phi \le \frac{\pi}{2}$, while being below the cone gives $\cos\phi \le \sin\phi$, or $\phi \ge \frac{\pi}{4}$. Combining these restrictions gives $\frac{\pi}{4} \le \phi \le \frac{\pi}{2}$. Thus,

$$E = \left\{(\rho, \theta, \phi) : \frac{\pi}{4} \le \phi \le \frac{\pi}{2},\ 0 \le \theta \le 2\pi,\ 0 \le \rho \le 2\right\}.$$

The volume is

$$\begin{aligned}
\text{Vol}(E) &= \int_{\pi/4}^{\pi/2} \int_0^{2\pi} \int_0^2 \rho^2 \sin\phi \, d\rho \, d\theta \, d\phi \\
&= \frac{8}{3} \int_{\pi/4}^{\pi/2} \int_0^{2\pi} \sin\phi \, d\theta \, d\phi \\
&= \frac{16\pi}{3} \left[-\cos\phi\right]_{\pi/4}^{\pi/2} \\
&= \frac{16\pi}{3} \left(0 + \frac{\sqrt{2}}{2}\right) \\
&= \frac{8\sqrt{2}\pi}{3}.
\end{aligned}$$

7. Use spherical coordinates to find the volume of the solid in the first octant that lies inside the sphere $x^2 + y^2 + z^2 = 9$ and above the plane $z = 1$.

In spherical coordinates, the sphere becomes $\rho = 3$, and the plane $z = 1$ becomes $\rho\cos\phi = 1$, or $\rho = \sec\phi$. Since the solid lies in the first octant, we have $0 \le \theta \le \frac{\pi}{2}$ and $0 \le \phi \le \frac{\pi}{2}$. Along a ray from the origin that meets the solid, ρ runs from $\sec\phi$ at the plane to 3 at the sphere. The sphere and plane intersect when $\sec\phi = 3$, and a ray meets the solid only if $\sec\phi \le 3$, so $0 \le \phi \le \arccos(1/3)$. Thus,

$$E = \left\{(\rho, \theta, \phi) : 0 \le \phi \le \arccos\left(\frac{1}{3}\right),\ 0 \le \theta \le \frac{\pi}{2},\ \sec\phi \le \rho \le 3\right\}.$$

The volume is

$$\begin{aligned}
\text{Vol}(E) &= \int_0^{\arccos(1/3)} \int_0^{\pi/2} \int_{\sec\phi}^3 \rho^2 \sin\phi \, d\rho \, d\theta \, d\phi \\
&= \frac{1}{3} \int_0^{\arccos(1/3)} \int_0^{\pi/2} (27 - \sec^3\phi) \sin\phi \, d\theta \, d\phi \\
&= \frac{\pi}{6} \int_0^{\arccos(1/3)} (27 - \sec^3\phi) \sin\phi \, d\phi.
\end{aligned}$$

For the second term, let $u = \cos\phi$, so $du = -\sin\phi \, d\phi$. Then

$$\int -\sec^3\phi \sin\phi \, d\phi = \int u^{-3} \, du = -\frac{1}{2u^2} + C = -\frac{1}{2\cos^2\phi} + C.$$

Therefore,

$$\begin{aligned}\mathrm{Vol}(E) &= \frac{\pi}{6}\left[-27\cos\phi - \frac{1}{2\cos^2\phi}\right]_0^{\arccos(1/3)} \\ &= \frac{\pi}{6}\left[\left(-9-\frac{9}{2}\right)-\left(-27-\frac{1}{2}\right)\right] \\ &= \frac{\pi}{6}(14) \\ &= \frac{7\pi}{3}.\end{aligned}$$

8. Evaluate the following integral by switching to spherical coordinates.

$$\int_{-2}^{2}\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}}\int_{2-\sqrt{4-x^2-y^2}}^{2+\sqrt{4-x^2-y^2}} (x^2+y^2+z^2)^{3/2}\,dz\,dy\,dx$$

The bounds describe the ball $x^2+y^2+(z-2)^2 \le 4$, which is centered at $(0,0,2)$ and has radius 2. In spherical coordinates, its boundary becomes

$$\rho^2 - 4\rho\cos\phi = 0 \quad\Longleftrightarrow\quad \rho(\rho - 4\cos\phi) = 0.$$

The ball lies above the xy-plane, so $0 \le \phi \le \frac{\pi}{2}$. Along each such ray, ρ runs from 0 to $4\cos\phi$. Also, $(x^2+y^2+z^2)^{3/2} = \rho^3$. Including the factor $\rho^2\sin\phi$ from the change of coordinates, we obtain

$$\begin{aligned}\int_0^{\pi/2}\int_0^{2\pi}\int_0^{4\cos\phi} \rho^5\sin\phi\,d\rho\,d\theta\,d\phi &= \frac{4096}{6}\int_0^{\pi/2}\int_0^{2\pi}\cos^6\phi\sin\phi\,d\theta\,d\phi \\ &= \frac{4096\pi}{3}\int_0^{\pi/2}\cos^6\phi\sin\phi\,d\phi \\ &= \frac{4096\pi}{3}\left[-\frac{\cos^7\phi}{7}\right]_0^{\pi/2} \\ &= \frac{4096\pi}{21}.\end{aligned}$$

SOME USEFUL DEFINITIONS, THEOREMS, AND NOTATION

Cylindrical Coordinates. In the cylindrical coordinate system, a point P in space is represented by an ordered triple (r, θ, z), where (r, θ) gives the polar coordinates of the projection of P onto the xy-plane, and z is the same z-coordinate as in rectangular coordinates:

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad z = z.$$

If a solid E can be described in cylindrical coordinates by

$$E = \{(r, \theta, z) \mid \alpha \le \theta \le \beta,\ h_1(\theta) \le r \le h_2(\theta),\ u_1(r, \theta) \le z \le u_2(r, \theta)\},$$

then

$$\iiint_E f(x, y, z)\, dV = \int_\alpha^\beta \int_{h_1(\theta)}^{h_2(\theta)} \int_{u_1(r,\theta)}^{u_2(r,\theta)} f(r\cos\theta, r\sin\theta, z)\, r\, dz\, dr\, d\theta.$$

Cylindrical coordinates can also be defined using polar coordinates in the yz-plane or the xz-plane instead of the xy-plane, and in each case the Jacobian factor is still r.

Spherical Coordinates. In the spherical coordinate system, a point P in space is represented by an ordered triple (ρ, θ, ϕ), where ρ is the distance from the origin to P, θ is the same angle as in cylindrical coordinates, and ϕ is the angle between the positive z-axis and the line segment from the origin to P:

$$x = \rho\sin\phi\cos\theta, \qquad y = \rho\sin\phi\sin\theta, \qquad z = \rho\cos\phi.$$

If a solid E can be described in spherical coordinates by

$$E = \{(\rho, \theta, \phi) \mid a \le \phi \le b,\ h_1(\phi) \le \theta \le h_2(\phi),\ u_1(\theta, \phi) \le \rho \le u_2(\theta, \phi)\},$$

then

$$\iiint_E f(x, y, z)\, dV = \int_a^b \int_{h_1(\phi)}^{h_2(\phi)} \int_{u_1(\theta,\phi)}^{u_2(\theta,\phi)} f(\rho\sin\phi\cos\theta, \rho\sin\phi\sin\theta, \rho\cos\phi)\rho^2\sin\phi\, d\rho\, d\theta\, d\phi.$$

Suggested Textbook Problems
Section 15.7: 1-32
Section 15.8: 1-32,37-40,43-46